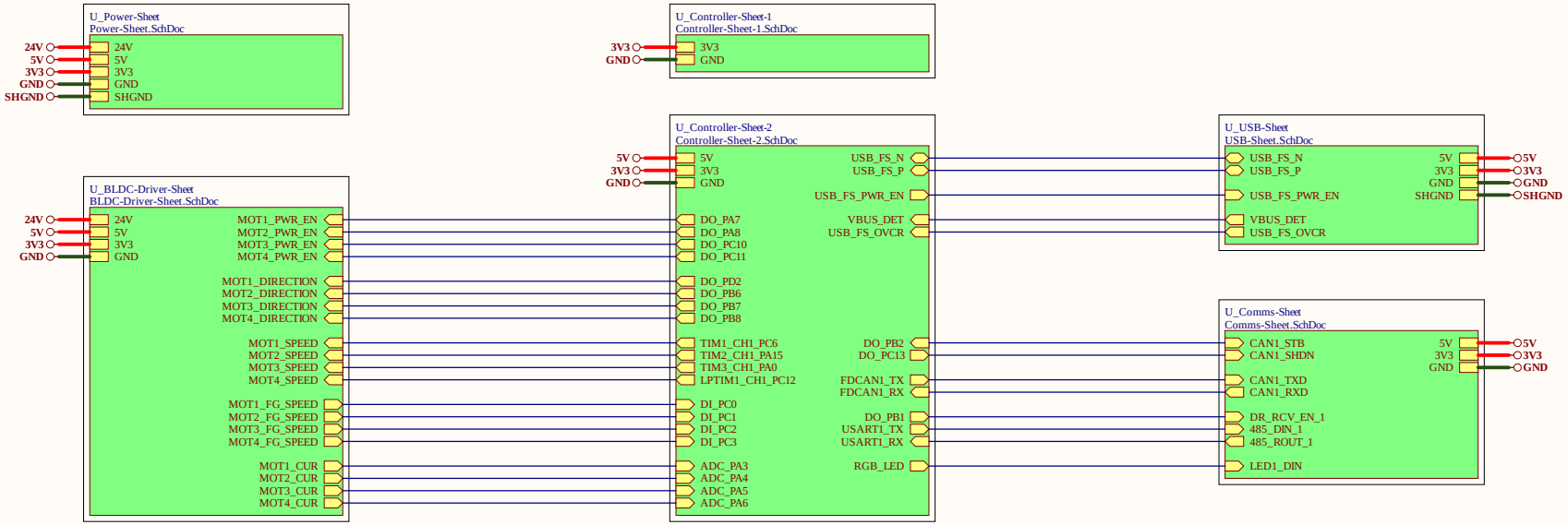
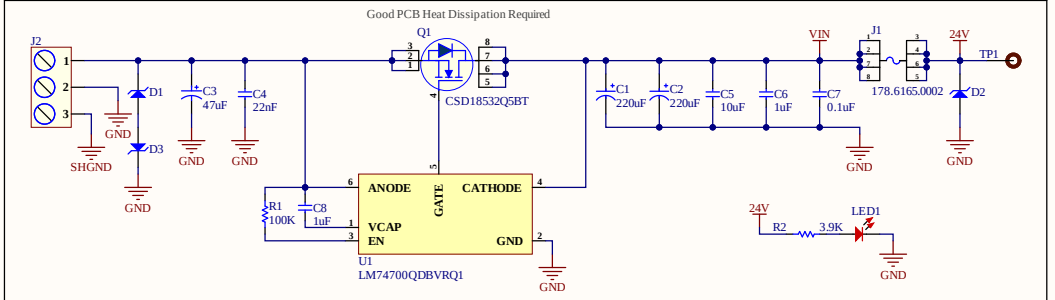
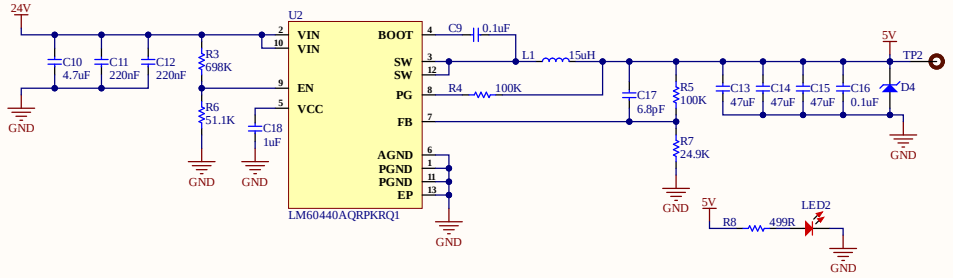


Title: Top-Sheet.SchDoc	
File: BLDC-Driver-v2.0.PrjPcb	
Date: 15/01/2025	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 7

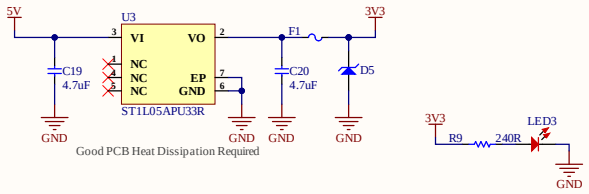
24V INPUT

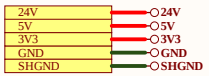


5V BUCK CONVERTER



3.3V LINEAR REGULATOR

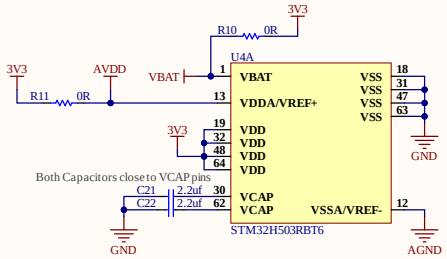




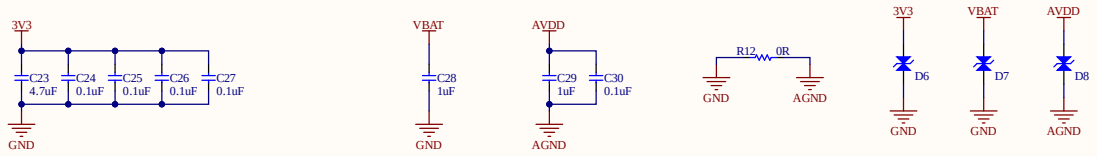
eSpark Consulting Group

Title: Power-Sheet.SchDoc		
File: BLDC-Driver-v2.0.PrjPcb		
Date: 15/01/2025	Page	Num: 2
Drawn By: Huzaifa Ali	Page	Total: 7

CONTROLLER POWER SHEET



MCU DECOUPLING CAPACITORS

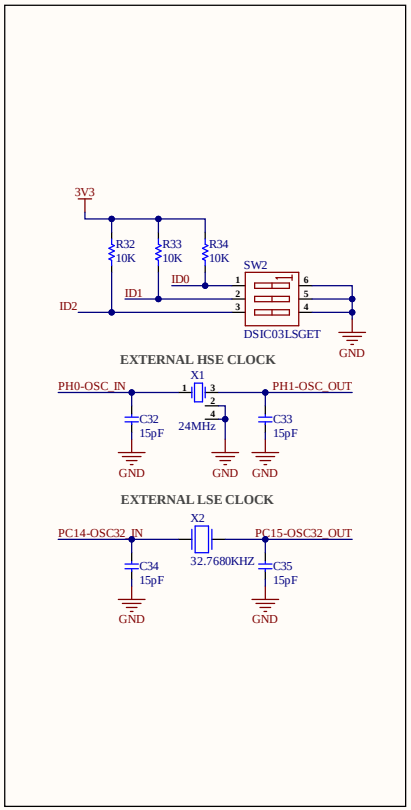


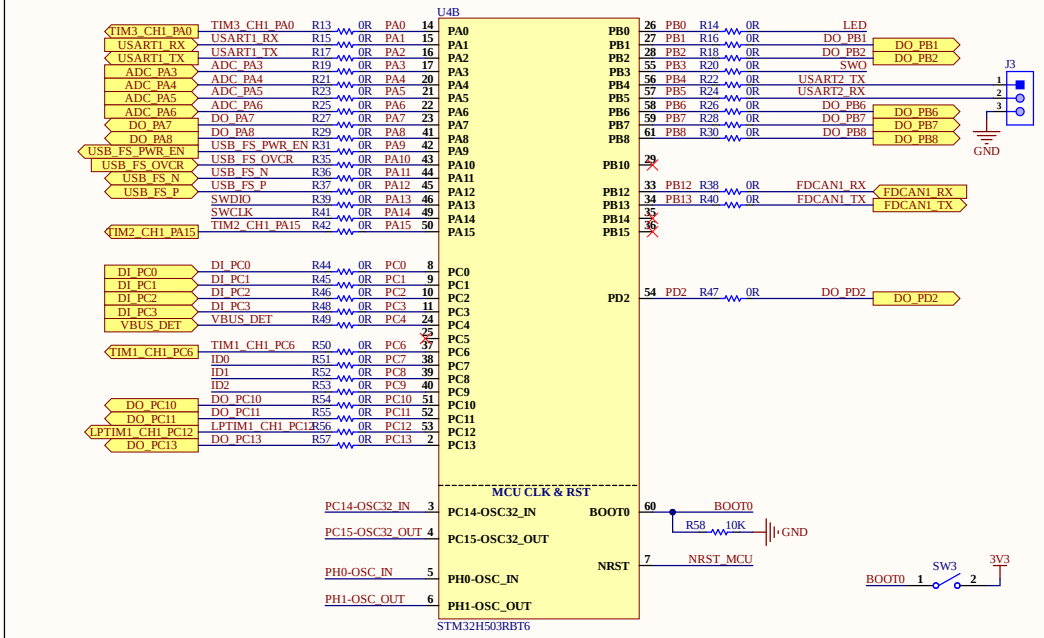


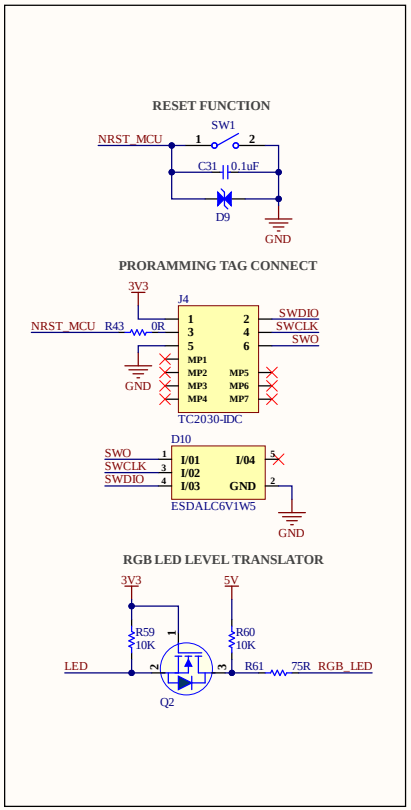
eSpark Consulting Group

Title: Controller-Sheet-1.SchDoc		
File: BLDC-Driver-v2.0.PrjPcb		
Date: 15/01/2025	Page	Num: 3
Drawn By: Huzaifa Ali	Page	Total: 7

CONTROLLER IO SHEET







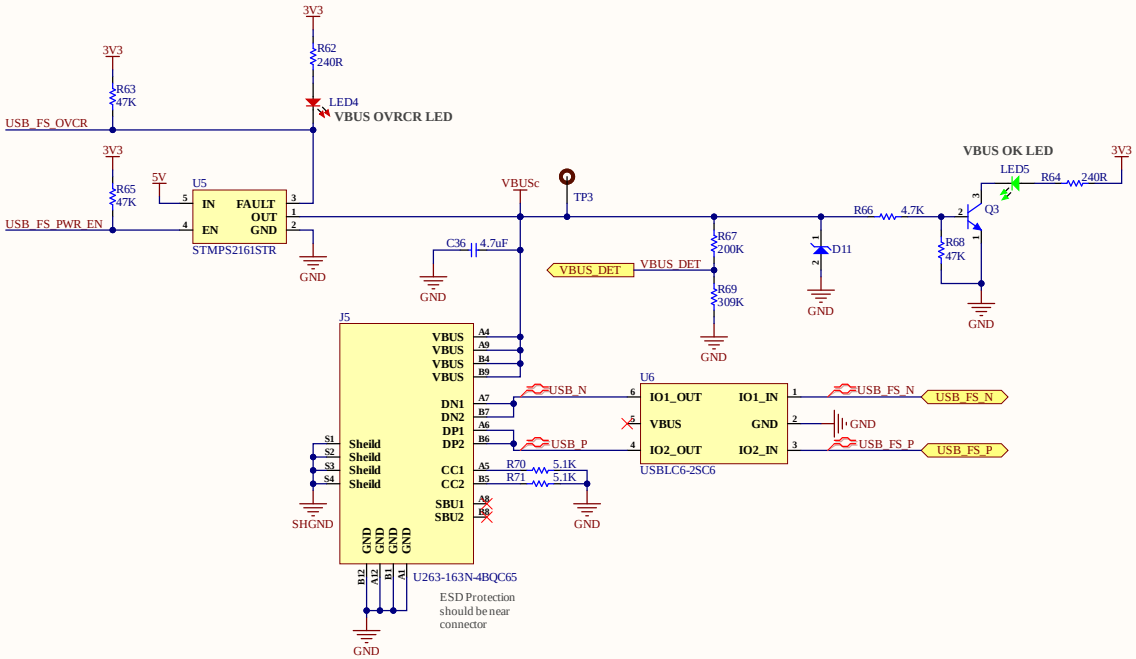
RGB_LED — RGB_LED

5V
3V3
GND

Title:	Controller-Sheet-2.SchDoc
File:	BLDC-Driver-v2.0.PrjPcb
Date: 15/01/2025	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 7

eSpark Consulting Group

USB



USB_FS_PWR_EN
USB_FS_OVCR

USB_FS_PWR_EN
USB_FS_OVCR

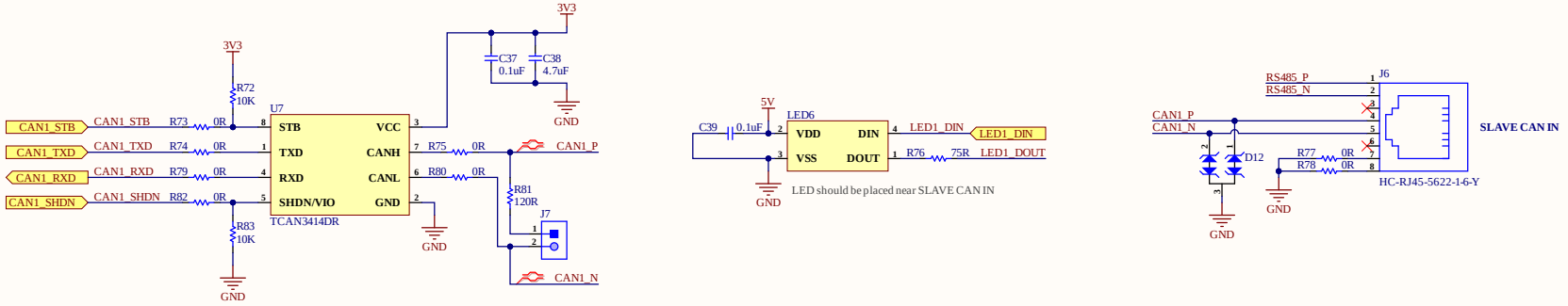
5V
3V3
GND
SHGND

5V
3V3
GND
SHGND

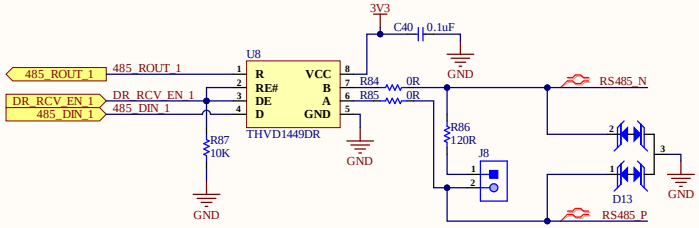
eSpark Consulting Group

Title: USB-Sheet.SchDoc		
File: BLDC-Driver-v2.0.PrjPcb		
Date: 15/01/2025	Page	Num: 5
Drawn By: Huzaifa Ali	Page	Total: 7

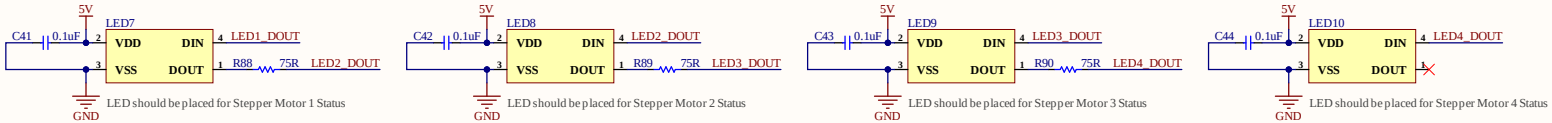
CAN



RS485



LEDs

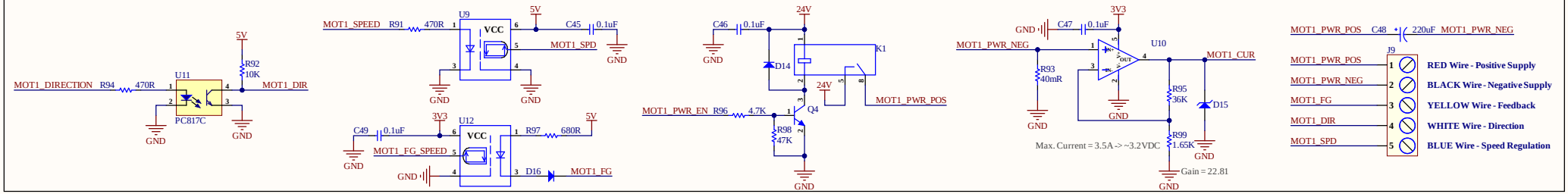




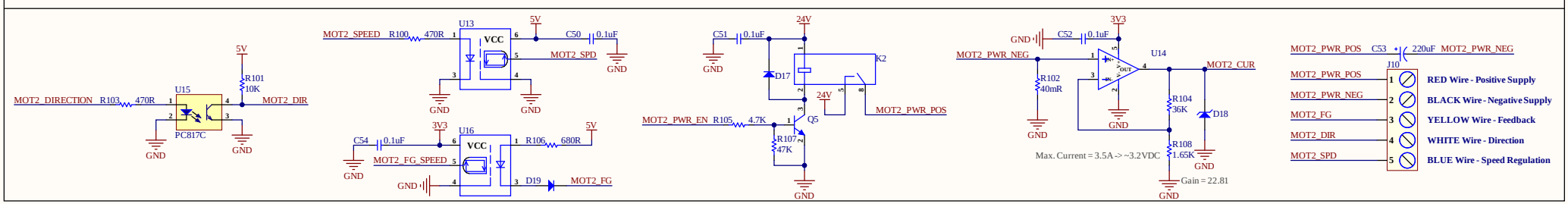
eSpark Consulting Group

Title: Comms-Sheet.SchDoc		
File: BLDC-Driver-v2.0.PrjPcb		
Date: 15/01/2025	Page	Num: 6
Drawn By: Huzaifa Ali	Page	Total: 7

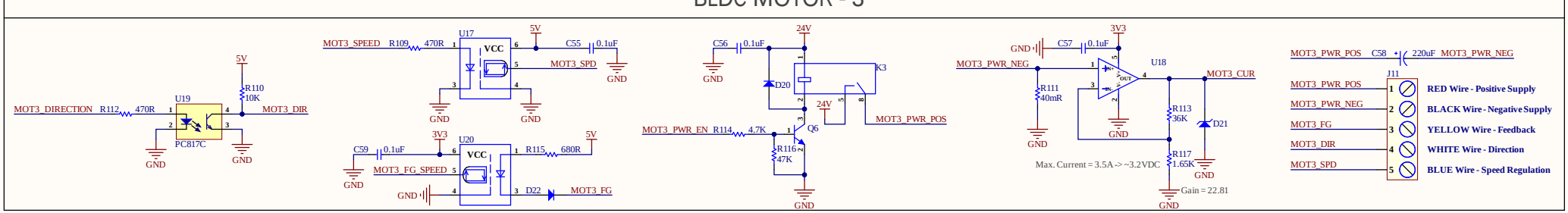
BLDC MOTOR - 1



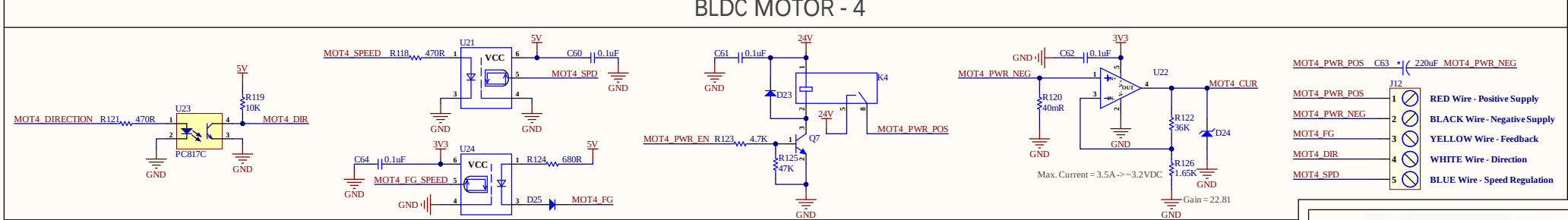
BLDC MOTOR - 2



BLDC MOTOR - 3



BLDC MOTOR - 4



MOT1_DIRECTION MOT1_DIRECTION
MOT2_DIRECTION MOT2_DIRECTION
MOT3_DIRECTION MOT3_DIRECTION
MOT4_DIRECTION MOT4_DIRECTION

MOT1_SPEED MOT1_SPEED
MOT2_SPEED MOT2_SPEED
MOT3_SPEED MOT3_SPEED
MOT4_SPEED MOT4_SPEED

MOT1_FG_SPEED MOT1_FG_SPEED
MOT2_FG_SPEED MOT2_FG_SPEED
MOT3_FG_SPEED MOT3_FG_SPEED
MOT4_FG_SPEED MOT4_FG_SPEED

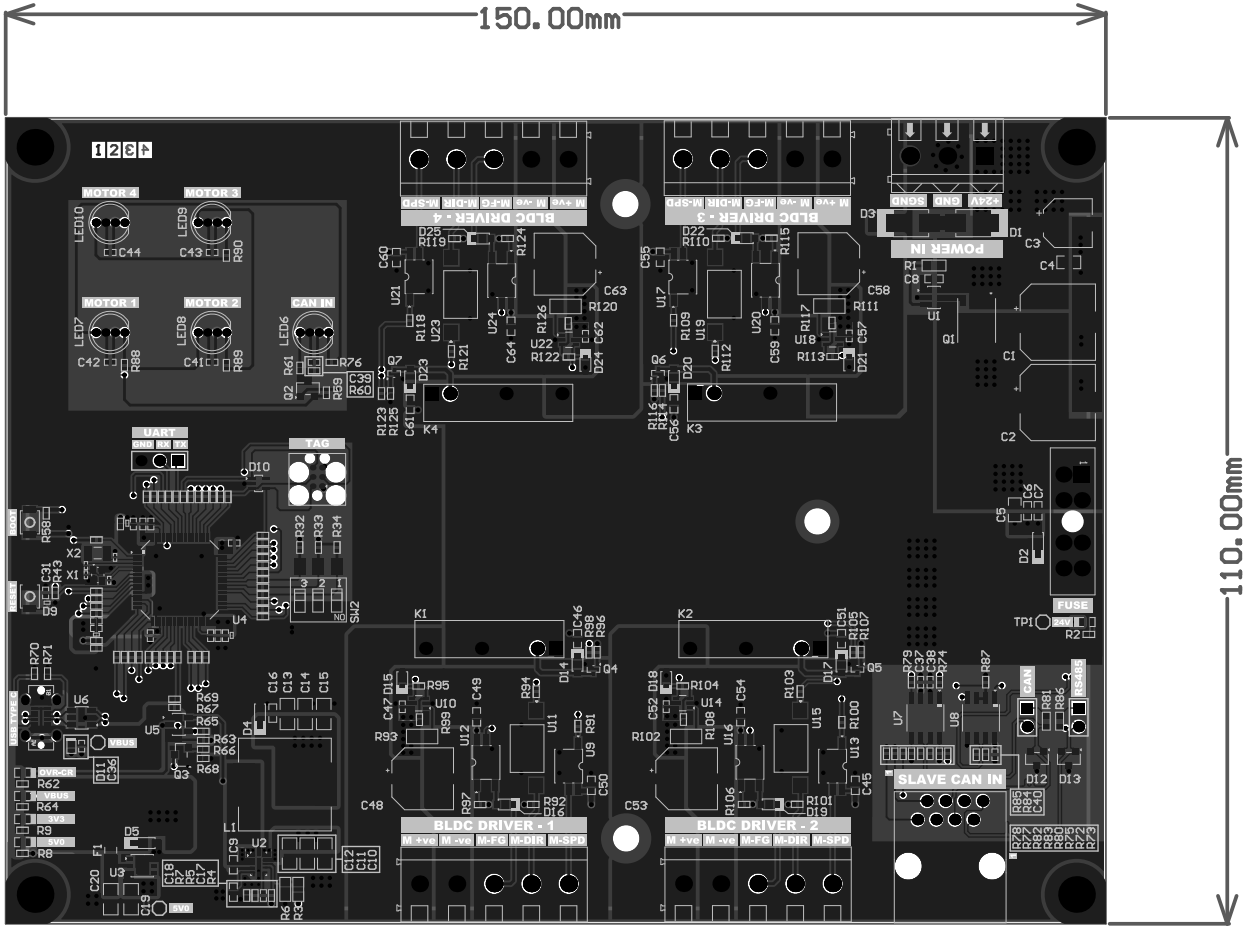
MOT1_PWR_EN MOT1_PWR_EN
MOT2_PWR_EN MOT2_PWR_EN
MOT3_PWR_EN MOT3_PWR_EN
MOT4_PWR_EN MOT4_PWR_EN

MOT1_CUR MOT1_CUR
MOT2_CUR MOT2_CUR
MOT3_CUR MOT3_CUR
MOT4_CUR MOT4_CUR

24V 24V
5V 5V
3V3 3V3
GND GND

eSpark Consulting Group

Title: BLDC-Driver-Sheet.SchDoc	
File: BLDC-Driver-v2.0.PrjPcb	
Date: 15/01/2025	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 7



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.010mm	3.5	
1	Top Layer		0.035mm		
	Dielectric 2	PP-006	0.203mm	4.4	
2	Layer 1	CF-004	0.035mm		
	Dielectric 1	FR-4	1.030mm	4.6	
3	Layer 2	CF-004	0.035mm		
	Dielectric 3	PP-006	0.203mm	4.4	
4	Bottom Layer		0.035mm		
	Bottom Solder	Solder Resist	0.010mm	3.5	
	Bottom Overlay				
Total board thickness:			1.596mm		